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RCED v5.1 — Robustness to shocks

A 30-year stress test of the three regimes: random exogenous shocks (Monte-Carlo, 600 trajectories), with the same shock sequence applied to all three regimes.

RCED v5.1 | Social-democratic | Free market

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Illustrative parametric model. Robustness is measured by a system stress S (S ≤ 1 = within the 9 planetary boundaries) subjected to random shocks. Differences come only from each regime's response, the shock sequence being identical. Read as a comparison of dynamics, not as an empirical forecast.

1. Purpose & method

This simulation extends the three-regime comparison to where the 20-year projection stopped: **durability under shocks**. Each regime is subjected to the same sequence of random exogenous shocks (energy, climate, financial, political crises) over 30 years, repeated across 600 Monte-Carlo trajectories.

The system state is summarised by a **stress S(t)** representing pressure on the most binding planetary boundary: **S ≤ 1 = all 9 boundaries respected**, S > 1 = breach. A shock adds an impulse to S; the system then returns toward its equilibrium at a speed specific to each regime. IVB tracks the stress: it falls when S rises.

> **A FAIR DESIGN**
>
> The **same** shock sequence hits all three regimes in each trajectory: the robustness gap therefore does not come from luck, but from each model's *structural response*. The parameters are stylised (§ 7) and calibrated to the documented regimes.

2. Robustness mechanism — three levers

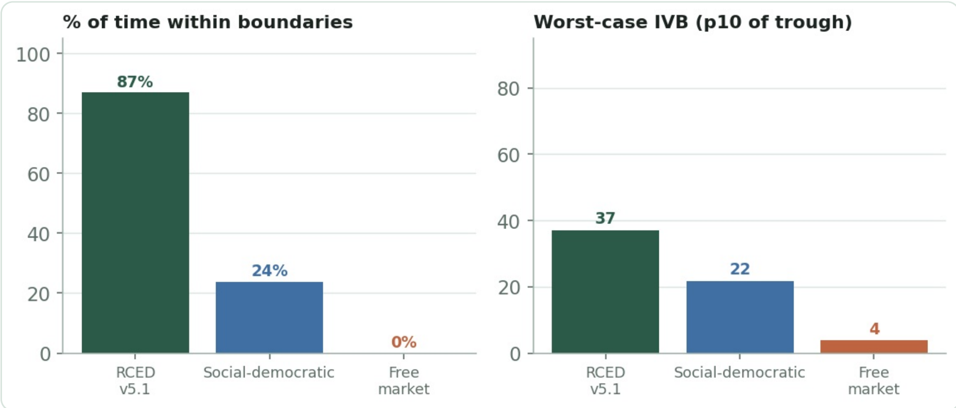
Three structural properties determine how each regime holds up under shocks:

Lever	RCED v5.1	Social-democratic	Free market
Structural margin (equilibrium S*)	0.55 — wide buffer	0.86 — thin margin	1.25 — already in breach
Restoring force (return to equilibrium)	strong (half-life 1.26 yr)	medium (half-life 1.54 yr)	weak (half-life 3.85 yr)
Political reversibility	none — constitutional ceiling (F8/F9)	deregulation risk under crisis (ratchet)	no protection
Effect of a shock	absorbed below the limit	breach, slow return + erosion	amplified and persistent

RCED's use-property removes the incentive to rent and overproduce: equilibrium sits well below the limit (buffer), and the restoring force is structural (not dependent on political will). Social democracy holds via regulation, grafted onto an intact engine: thin and reversible margin. The market has neither buffer nor restoring force.

3. Results — robustness metrics

Metric (30 years, 600 runs)	RCED v5.1	Social-democratic	Free market
% of time within the 9 boundaries	**87%**	**24%**	**0%**
Median IVB at year 30	**74.9**	**53.5**	**13.0**
Worst-case IVB (p10 of trough)	**37.1**	**21.8**	**3.9**
Recovery half-life	**1.26 yr**	**1.54 yr**	**3.85 yr**
Breach episodes (/30 years)	**3.6**	**2.4**	**0.0**



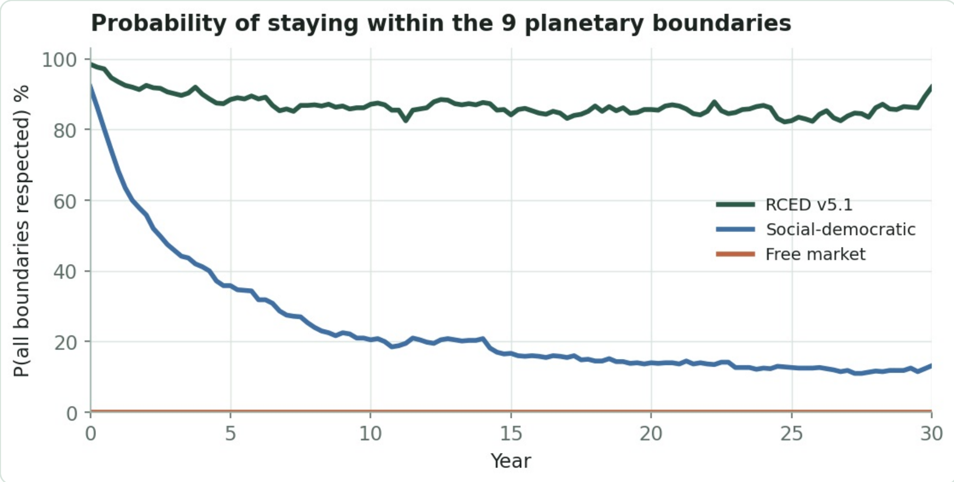
Left: share of time spent within the 9 boundaries. Right: worst-case IVB (10th percentile of each trajectory's trough).

READING THE MARKET'S "0 BREACHES"

The market shows 0 breach episodes not because it never breaches, but because it **starts already in breach** (S* = 1.25) and never leaves: there is therefore no "entry" into breach to count. Its time within the boundaries is 0%.

4. Probability of staying within the boundaries

This is the most telling result. RCED stabilises around **87%** probability of respecting the 9 boundaries, shock after shock — its margin absorbs most impulses. Social democracy starts high (it reaches the zone) but **erodes** toward ~24%: thin margin + deregulation ratchet. The market stays at 0%.

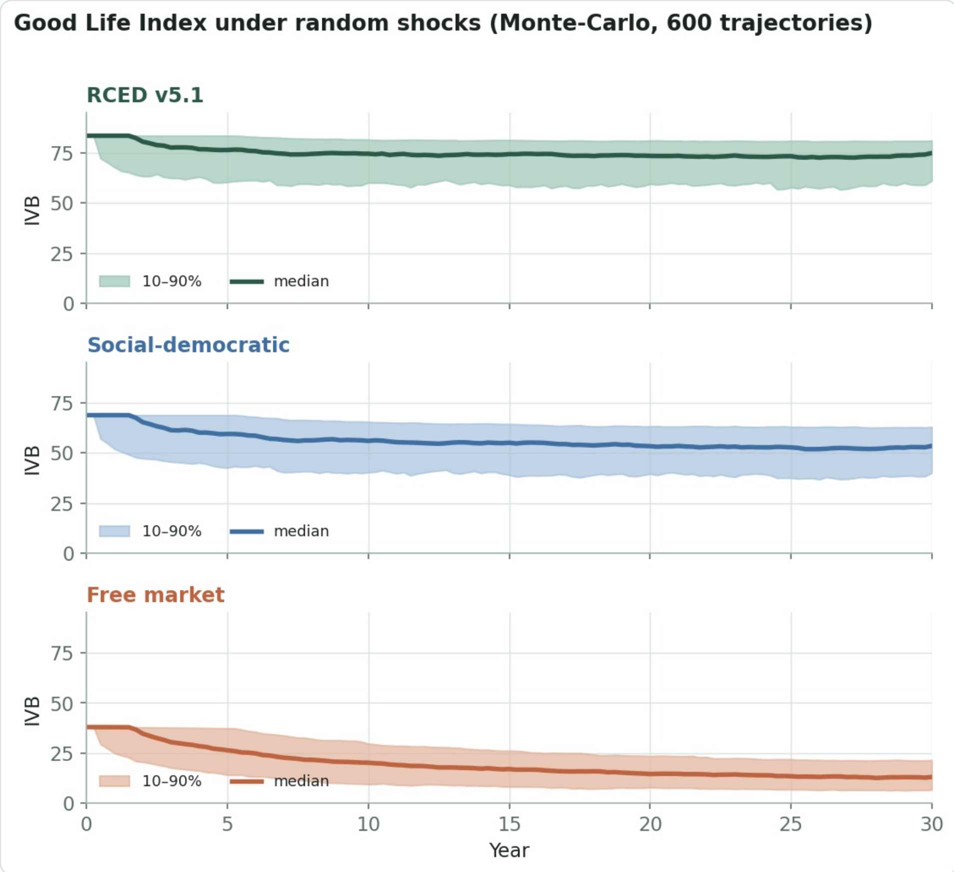


Probability, at each date, that all 9 boundaries are simultaneously respected (over 600 trajectories).

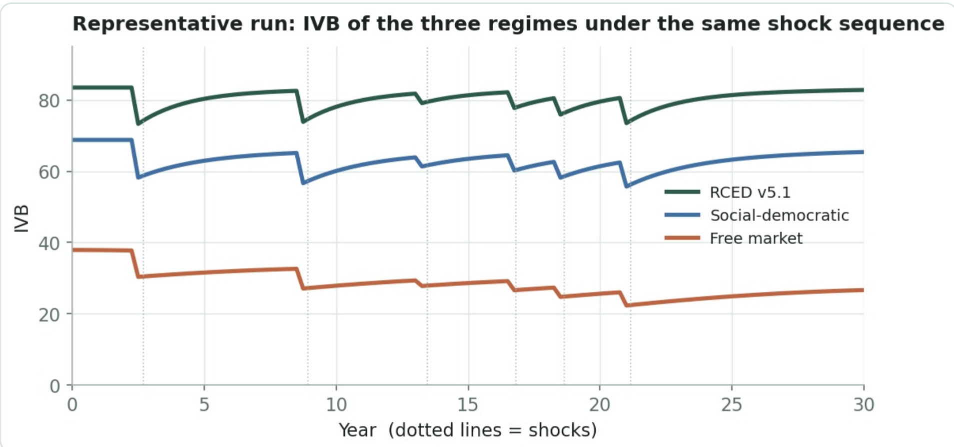
The decline of the social-democratic curve illustrates the "long-run durability" the 20-year simulation could not capture: a system that reaches the zone does not necessarily stay there once shocks accumulate and regulation can be unwound.

5. Good Life Index under shocks

In the Monte-Carlo distribution, RCED's IVB stays high and tight (narrow band: predictable), social democracy's drifts downward with a wider band (uncertain), the market's collapses.



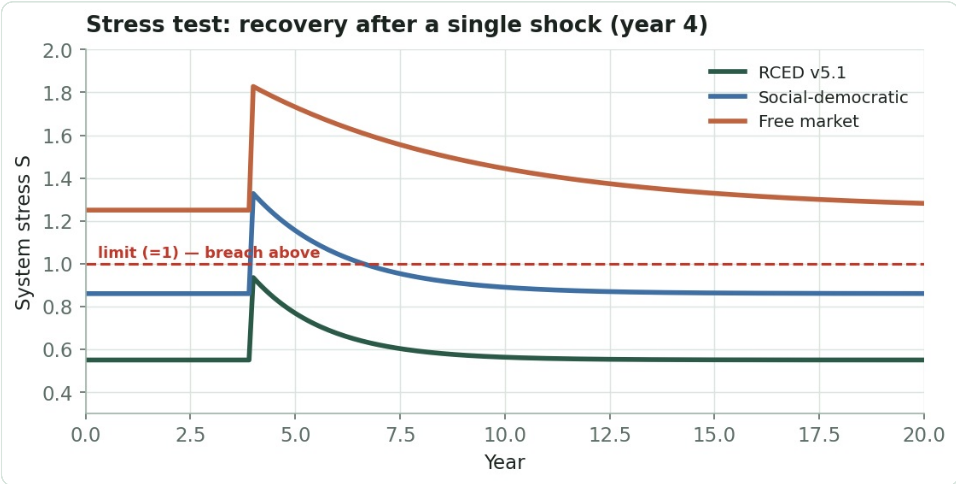
IVB under shocks: median and 10–90% band over 600 trajectories, by regime.



Representative run: IVB of the three regimes under the same shock sequence (dotted lines). RCED takes the hit and rebounds; social democracy slips; the market stays low.

6. Stress test — recovery after a shock

A single standardised shock (year 4) reveals two properties at once: the **margin** (how high S rises) and the **recovery speed**. RCED, starting at 0.55, does not even cross the limit (1.26 yr half-life). Social democracy crosses the limit and takes ~3 years to drop back below it. The market, already above, never returns.



Recovery of stress S after a single shock. The red line (S = 1) separates the safe zone from breach.

7. Reading & limits of the exercise

WHAT THE SIMULATION CONFIRMS

The robustness ordering (87% / 24% / 0% of time within the boundaries) follows from three structural properties, not from ad-hoc assumptions about a regime's "virtue": **margin**, **restoring force**, **irreversibility**. This is the model's central § 5.3 argument, made dynamic: structurally removing the rent engine confers a durability that grafted regulation does not match.

LIMITS

- **Stylised model**: the stress S is a one-dimensional aggregate; real boundaries interact non-linearly.
- The social-democratic "deregulation ratchet" is an assumption (political reversibility under crisis), plausible but unproven; without it, social democracy would hold up better.
- The regimes face *identical* shocks; in reality a sober regime is also less *exposed* to some shocks (energy, supply chains) — an advantage not fully credited here.
- Calibration of margins, restoring forces and shock frequency (~1 every 2.5 years) is set by the author; the levels depend on these choices, the regime *ordering* is robust to wide variations.

A. Simulation equations

Stress dynamics (per regime, step dt = 0.25 yr)

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S(t+dt) = S(t) + κ · ( S* + residual − S(t) ) · dt + Σ shocks(t)·sensitivity
viable(t) ⇔ S(t) ≤ 1 (all 9 boundaries respected)
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κ = restoring force ; S* = structural equilibrium ; residual = persistent part of shocks (fades slowly).

Shocks & reversibility

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shocks: Poisson process (λ = 0.4/yr) · magnitude ~ Gamma (mean ≈ 0.36)
social-democratic: at each shock, prob. p_rev → S* rises by δ (deregulation ratchet)
market: slow drift of S* (+0.012/yr) + large persistent share (amplification)
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Coupling to IVB

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IVB(t) = IVB* · exp( −γ · max(0, S(t) − S*₀ ) ), γ = 0.75
IVB* : RCED 83.5 · social-democratic 68.8 · market 37.9 (regimes' Y20 values)
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Per-regime parameters (S^*_0 ; κ ; sensitivity)

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RCED v5.1 : 0.55 ; 0.55 ; 0.70 — wide margin, strong restoring, no ratchet
Social-democratic : 0.86 ; 0.45 ; 0.85 — thin margin, ratchet p_rev = 0.15 (δ = 0.045)
Free market : 1.25 ; 0.18 ; 1.05 — in breach, weak restoring, drift
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